

Business Requirements Document

HealthConnect Patient Portal

Client: Meridian Health Systems

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Prepared by: Business Analysis Team

Platform: Salesforce Health Cloud + Experience Cloud

1 Executive Summary

Meridian Health Systems requires a patient engagement portal built on Salesforce Health Cloud and Experience Cloud. The portal will allow patients to view appointments, access medical records, communicate with care teams, request prescription refills, and manage their care plans. The system must integrate with the existing Epic EHR system for real-time patient data synchronization.

The solution must comply with HIPAA regulations and support 50,000+ active patients with role-based access for patients, caregivers, and healthcare providers.

2 Functional Requirements

2.1 Patient Registration and Profile Management

REQ-001 The system shall allow patients to self-register on the portal using their email address, date of birth, and medical record number for identity verification.

REQ-002 The system shall automatically match self-registered patients with existing Person Account records in Salesforce using the medical record number as the unique identifier.

REQ-003 The system must provide a patient profile page where patients can view and update their contact information, emergency contacts, insurance details, and communication preferences.

REQ-004 The system shall capture patient consent for data sharing and communication electronically, storing consent records with timestamps and IP addresses for HIPAA compliance.

REQ-005 Users should be able to upload identity documents (driver's license, insurance card) directly from the portal, with files stored securely in Salesforce Files.

2.2 Appointment Management

REQ-006 The system shall display a calendar view of all upcoming and past appointments, synced in real-time from the Epic EHR system via REST API integration.

REQ-007 Patients must be able to request new appointments by selecting a provider, specialty, preferred date range, and visit type through a guided screen flow.

REQ-008 The system shall send automated appointment reminders via email and SMS at 72 hours, 24 hours, and 2 hours before the scheduled appointment.

REQ-009 The system must allow patients to cancel or reschedule appointments up to 24 hours before the scheduled time, with cancellation reasons captured for analytics.

REQ-010 When an appointment is cancelled, the system shall automatically notify the care team and update the appointment status in both Salesforce and Epic EHR.

2.3 Medical Records and Care Plans

REQ-011 The system shall provide read-only access to patient medical records including lab results, imaging reports, and clinical notes, retrieved from Epic EHR via HL7 FHIR API.

REQ-012 The system must display active care plans with goals, tasks, and progress tracking. Patients should be able to mark care plan tasks as completed.

REQ-013 The system shall calculate and display a Patient Health Score based on care plan adherence, appointment attendance, and medication compliance, using a custom formula.

REQ-014 Lab results flagged as critical must trigger an immediate notification to both the patient and their assigned care coordinator via Platform Events.

REQ-015 The system should allow patients to download their medical records as PDF documents for personal use or sharing with external providers.

2.4 Secure Messaging

REQ-016 The system shall provide a HIPAA-compliant secure messaging feature allowing patients to communicate with their care team directly through the portal.

REQ-017 Messages must be stored as Activity records associated with the patient's Person Account, with encryption at rest and in transit.

REQ-018 The system shall support file attachments in messages (lab results, photos) with a maximum file size of 25MB and virus scanning before storage.

REQ-019 When a patient sends a message, the system must automatically route it to the appropriate care team member based on the message category (billing, medical, pharmacy, general) using assignment rules.

2.5 Prescription Management

REQ-020 The system shall display a list of active and past prescriptions retrieved from the pharmacy module in Epic EHR.

REQ-021 Patients must be able to request prescription refills through the portal, with the request routed to the assigned physician for approval via a Flow-based approval process.

REQ-022 The system shall track prescription refill status (Requested, Under Review, Approved, Denied, Fulfilled) with real-time status updates visible to the patient.

REQ-023 The system must send automated notifications when a prescription is due for refill within 7 days, calculated based on the prescription fill date and day supply.

3 Non-Functional Requirements

3.1 Security and Compliance

NFR-001 The system must enforce multi-factor authentication for all patient portal users using Salesforce MFA capabilities.

NFR-002 All patient health information must be encrypted at rest using Salesforce Shield Platform Encryption and in transit using TLS 1.3.

NFR-003 The system shall implement role-based access control with separate profiles for Patients, Caregivers, Care Coordinators, Physicians, and Portal Administrators.

NFR-004 The system must maintain a complete audit trail of all data access and modifications to patient records using Salesforce Field Audit Trail.

NFR-005 Patient data visibility must be restricted so that patients can only see their own records, caregivers can see their assigned patients, and providers can see patients in their care team.

3.2 Performance and Scalability

NFR-006 The portal must support concurrent access by at least 5,000 users with page load times under 3 seconds.

NFR-007 The Epic EHR integration must process appointment sync updates within 30 seconds of the source system change.

4 Reporting Requirements

REQ-024 The system shall provide a Patient Engagement Dashboard showing appointment adherence rates, portal login frequency, message response times, and care plan completion percentages.

REQ-025 The system must generate monthly Patient Satisfaction Reports with drill-down by department, provider, and visit type, accessible to hospital administrators.

REQ-026 Care coordinators should have a real-time dashboard showing their assigned patients' health scores, upcoming appointments, and pending tasks.

5 Integration Requirements

REQ-027 The system shall integrate bidirectionally with Epic EHR via REST API for patient demographics, appointments, lab results, and prescription data with a sync frequency of every 15 minutes.

REQ-028 The system must integrate with Twilio for SMS notifications and appointment reminders.

REQ-029 The system shall integrate with DocuSign for electronic consent form signing and storage.

REQ-030 The system must provide a webhook endpoint for Epic EHR to push real-time critical lab result notifications.